# …… Massive Crowd Management in Urban Pedestrian Zones

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***Abstract*—This paper presents a bi-level optimization framework for crowd management in large pedestrian zones, integrating anisotropic geometric guidance with discrete directional constraints. The lower level employs a Hughes-type macroscopic crowd model enhanced with metric tensor fields for channel guidance and constrained Hamilton-Jacobi-Bellman (HJB) equations for one-way channel control. The upper level optimizes channel direction configurations and guidance intensities using a Structure-Aware Hybrid Block Optimization (SA-HBO) algorithm. Three raw physical metrics are introduced for efficiency, safety, and load balancing, and are further converted into dimensionless indicators for cross-scheme comparison and scalarization. The proposed method exploits the local response characteristics of Bellman potential fields and employs surrogate pre-screening with Bellman warm-start strategies to reduce computational cost. Numerical experiments demonstrate the effectiveness of the proposed approach.**

***Index Terms*—Crowd dynamics, macroscopic modeling, channel optimization, Hamilton-Jacobi-Bellman equation, bi-level optimization.**

Fig. 1. Large-scale crowd at the Shanghai Nanjing East Road–Bund scenario during a holiday period.

## I. Introduction

LARGE-SCALE crowd gathering risks not only threaten individual safety but may also escalate into stampede incidents that disrupt social order and public security; the effectiveness of crowd management directly tests urban governance capacity and the modernization level of social safety governance systems [1], [2]. With the increasing frequency of high-density pedestrian activities such as holiday tourism, urban public events, commercial district promotions, New Year celebrations, and major sports events, core urban areas must accommodate pedestrian demands far exceeding daily levels within short time windows, posing enormous challenges for on-site organization, passage management, risk early warning, and emergency dispatch [3], [4], [5]. Against this backdrop, scientific modeling, numerical simulation, and optimization methods for formulating and evaluating crowd control strategies—improving the efficiency of large-scale event organization, reducing local congestion risks, and safeguarding public safety—have become critical scientific and technological problems demanding urgent breakthroughs in the fields of social safety and urban governance [6], [7].

Among various urban crowd gathering scenarios, open scenic areas, waterfront viewing platforms, historic commercial districts, and urban pedestrian corridors are highly representative. Unlike enclosed venues such as stadiums and subway stations with well-defined boundaries [8], [9], open urban scenic areas typically feature multiple entrances and exits, multi-stage touring, multi-directional weaving, and diverse route preferences [10]. Visitors do not simply move from one entrance to one exit; rather, they undergo a continuous behavioral chain: entering the scenic area, moving along the main tourist corridor, pausing for sightseeing, choosing passages to leave, and flowing back along external roads [11]. Consequently, crowd movement in such areas is governed not only by destination choice, passage accessibility, and congestion state, but also by spatial geometry, control regulations, and historical behavioral preferences [12], [11].

Taking the Shanghai Nanjing East Road–Bund area as an example, visitors typically approach the Bund from Nanjing East Road, ascend to the viewing platform via multiple stepped passages, tour along the waterfront direction, then exit through several southern or adjacent passages and return along lower streets or external roads. This scenario can be abstracted as a typical "waterfront platform–multi-stepped passage" crowd flow scenario, and further generalized as an "open scenic area with a main tourist corridor, multiple lateral connecting passages, and return flow lines." In such scenarios, management difficulties manifest in three aspects: first, multiple passages simultaneously serve entry, exit, and diversion functions, and local passage direction settings can alter overall flow organization; second, distinct phase-dependent behavioral transitions exist between the main tourist corridor and lateral passages, making single origin–destination models insufficient to capture realistic crowd behavior; third, local congestion and global route choice form mutual feedback loops, where control changes in one passage may trigger load redistribution in adjacent areas, thereby affecting overall system safety and efficiency. Fig. 1 shows the large-scale crowd at the Shanghai Nanjing East Road–Bund scenario during a holiday period.

Crowds in large-scale events constitute typical complex systems. Numerous individuals with heterogeneous desired speeds, destinations, route preferences, and reaction capabil-

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ities interact through multi-scale, nonlinear mechanisms in confined spaces, generating spatiotemporal patterns at the collective level [13], [4]. Macroscopically, these patterns manifest as destination-oriented flow, passage convergence, bottleneck queuing, local retention, exit competition, route diversion, and directional responses to control regulations [14], [15]; microscopically, individuals are influenced by surrounding density, visible paths, obstacle boundaries, guidance facilities, and herding behavior [16]. For open scenic areas, crowd behavior additionally exhibits pronounced phase dependence and stochasticity: visitors may first move toward the viewing platform, then tour along the main viewing direction, and subsequently choose different exit passages based on preferences or guidance [17]. These characteristics of multi-stage, multi-route coexistence with shared congestion mean that traditional static capacity analysis or single shortest-path models cannot adequately support refined management decisions.

In response to the complex characteristics of high-density crowd systems and the spatial organization features of open urban scenic areas, existing large-scale event security management practices typically integrate system dynamics theory, system coupling theory, and system safety theory, combined with the delineation of key high-risk zones, historical crowd behavior patterns, and event factors currently dominating crowd movement, to formulate a series of crowd behavior control measures [18], [19]. These measures include scientifically deploying security personnel at critical locations, erecting temporary barriers for effective isolation and diversion, regulating crowd direction and flow speed through manual guidance, implementing one-way passage on key channels, applying batch release in core areas, and promptly guiding visitors away from locally high-density zones, thereby enhancing overall safety assurance effectiveness [20], [21]. For instance, during major domestic holidays, the Shanghai Huangpu District Nanjing East Road–Bund area adopts measures such as batch release, passage direction regulation, south-entry north-exit routing, reduction of head-on conflicts, controlling inbound flow speed to the Bund through police deployment, and guiding lingering visitors to disperse [22]. Similarly, during peak periods, especially major holidays, Nanluoguxiang main street in Dongcheng District, Beijing, implements one-way passage measures, with visitors typically entering from the south entrance and exiting from the north [23]. Meanwhile, management deploys security and guidance personnel at key intersections along the main street, using loudspeakers, signage, and on-site guidance to direct visitors along prescribed routes, avoiding head-on conflicts and local congestion [23]. This is consistent in control mechanism with the "south-entry north-exit" and passage direction regulation employed in the Shanghai Bund area.

However, current management measures for urban high-pedestrian-flow areas still rely to a considerable extent on manual experience and static contingency plans. Although these measures hold significant value in practical management, deficiencies remain in scientific evaluation and refined optimization [24]. First, many control schemes are primarily formulated based on historical experience, lacking quantitative characterization of the impact of different passage direction configurations, geometric guidance intensities, and phase-dependent behavioral preferences. Second, existing methods tend to focus on single-point pedestrian flow statistics or local density monitoring, making it difficult to describe the dynamic coupling relationships among "passage direction rules–local optimal directions–collective density evolution–exit load distribution" [25]. Third, in open scenic area scenarios with multiple entrances, passages, and stages, neglecting visitor route preferences and phase transition behaviors can easily lead to misjudgment of passage loads and high-risk area locations. Finally, many existing simulation or optimization methods treat the crowd model simply as a black-box evaluator, failing to fully exploit the structural information among passage geometry, one-way rules, and multi-route subpopulations [26], resulting in insufficient efficiency in control strategy search [27] and hindering the interpretability of optimization results.

To address the above issues, this paper proposes a macroscopic crowd modeling and control optimization method incorporating one-way passage constraints, geometric guidance, and multi-stage route behavior, targeting the "waterfront platform–multi-stepped passage" open scenic area scenario. Specifically, this paper constructs a Bellman–conservation law coupled crowd continuum model: at the density evolution level, conservation laws describe the spatiotemporal propagation of subpopulations across different stages and routes; at the route choice level, independent potential functions are established for each stage–route subpopulation, with optimal directions computed via the discrete Bellman equation; at the congestion feedback level, subpopulations share the speed–density relationship determined by total density, thereby achieving mutual influence among multi-intent crowds. To represent passage direction control, one-way passage rules are formulated as constraints on the local admissible direction set, enabling the model to characterize how different configurations—such as one-way (forward/reverse) and two-way passage—affect local optimal directions. To represent passage geometric guidance, an anisotropic metric tensor is introduced to describe the cost differences of movement in different directions within traversable regions, thereby simulating pedestrian direction preferences and streamline organization when approaching, entering, and leaving stepped passages.

In terms of management optimization, this paper further distinguishes "controllable management variables" from "exogenous behavioral parameters." Passage direction configuration and geometric guidance intensity are variables that managers can regulate or design, while visitor route diversion preferences reflect the crowd's choice tendencies toward different routes or exits during stage transitions and should be identified from historical data, serving as fixed exogenous parameters input to the model during optimization, rather than being treated as variables freely controllable by managers. Based on this boundary, this paper adopts total travel time, high-density exposure time, and channel cumulative flow variance as efficiency, safety, and load balancing indicators, formulating a scalarized optimization problem under fixed weights to search over controllable variables. To improve optimization efficiency and enhance result interpretability, this paper designs a Structure-Aware Hybrid Block Optimization

framework (SA-HBO) and implements a simplified instance thereof. This method exploits the discrete neighborhood structure of passage direction variables, the continuous nature of geometric guidance variables, and the response characteristics of Bellman potential fields to local control changes, improving candidate strategy evaluation efficiency through surrogate pre-screening, full simulation confirmation, acceptance/rejection mechanisms, and Bellman warm-start procedures, while avoiding treating the model entirely as an unstructured black box.

The main contributions of this paper are as follows.

1) **A "waterfront platform–multi-stepped passage" crowd flow modeling framework for open scenic areas is proposed.** Targeting urban scenic area scenarios featuring a main tourist corridor, multiple lateral connecting passages, and return flow lines, this paper represents visitor behavior as a multi-stage, multi-route continuum flow process. The model allows different stage–route subpopulations to possess independent potential functions and target regions while sharing congestion feedback determined by total density, thereby capturing the coupled crowd evolution process involving destination orientation, route preferences, and local congestion.
2) **A Bellman–conservation law coupled model integrating one-way passage constraints and geometric guidance is constructed.** Passage direction control is represented as a local admissible direction set, allowing rules such as "forward," "reverse," and "bidirectional" to directly affect local optimal direction selection. Simultaneously, an anisotropic metric tensor is introduced to characterize the cost difference between axial and lateral movement within traversable regions, thereby simulating crowd behaviors such as pre-alignment, smooth convergence, and streamline reorganization near passage entrances. This modeling approach provides a mechanistic basis for explaining how control regulations alter local flow patterns.
3) **An evaluation framework distinguishing mechanism verification from strategy optimization is established.** Experimental indicators are categorized into behavioral mechanism indicators and management optimization indicators. The mechanism verification phase focuses on behavioral indicators such as direction consistency, counter-rule direction proportion, approach angle distribution, pre-alignment distance, and channel capture domain, used to examine whether the model can reproduce reasonable crowd flow behavior. The strategy evaluation and optimization phase employs total travel time, high-density exposure time, and channel cumulative flow variance to respectively measure efficiency, safety, and load balancing. This separation avoids the pitfall of substituting system performance metrics for model mechanism validation.
4) **A structure-aware optimization framework for controllable passage configuration is designed.** Under the condition of fixed historically identified preference parameters, this paper limits current optimization variables to passage direction configuration and geometric guidance intensity, and proposes a simplified instance of SA-HBO for solving the scalarized management optimization problem under fixed weights. Through discrete neighborhood search for passage direction configuration, continuous variable local search for geometric guidance intensity, combined with surrogate pre-screening, full simulation confirmation, best-so-far preservation, and acceptance/rejection mechanisms, the framework enhances the stability and interpretability of the optimization process.

The remainder of this paper is organized as follows. Section II reviews related work on massive crowd management and macroscopic crowd modeling. Section III presents the macroscopic crowd dynamics model, including the Bellman–conservation law coupled framework with one-way constraints and anisotropic metric tensor guidance. Section III-E describes the multi-stage, multi-route extension with probabilistic splitting. Section V defines the control evaluation metrics and the scalarized optimization objective. Section VI presents the SA-HBO optimization framework. Section VII reports numerical experiments covering mechanism validation, strategy comparison, and optimization results. Section VIII concludes the paper and discusses future work.

## II. Related Work

### A. Massive Crowd Management and Control

Crowd management in urban centers has been extensively studied from control-theoretic and optimization perspectives, aiming to enhance safety and efficiency through boundary regulation, facility layout, and intelligent decision-making [28], [29].

In control-theoretic approaches, Zhong et al. [30] designed boundary control strategies for urban traffic flow networks to ensure convergence to uncongested states under disturbances, with stability analysis based on Lyapunov methods [31]. For one-dimensional crowd dynamics, Wadoo and Kachroo [32] employed advection and diffusion control to prevent blocking and shock waves during evacuation. Addressing multi-directional disturbance propagation, Qin [33] proposed a unit sliding mode controller based on integral barrier Lyapunov functions to ensure boundedness of internal state variables, demonstrating strong disturbance rejection capabilities. For heterogeneous corridor regulation, Zhu et al. [34] combined policy learning with neural networks to learn optimal feedback controllers that adjust inflow rates and free-flow speeds, achieving balanced density distribution and reduced congestion.

Physical infrastructure regulation has also proven effective. Studies have shown that strategically placed obstacles can significantly improve outflow, velocity, and local density in merging areas [24], [35]. Yang et al. [36], [37] employed Model Predictive Control (MPC) to optimize barrier lengths at subway bottlenecks, later extending control variables to entrance limiters, gates, and escalator directions for dynamic, real-time crowd regulation. For evacuation guidance, Carmona and Paricio-Garcia [38] proposed dynamic exit-choice recommendations using multinomial Logit models. However,

their approach relies on discrete speed instructions (e.g., "stop," "walk," "run") transmitted via electronic wristbands [39], which poses implementation challenges in open urban scenarios.

Recently, intelligent optimization and deep reinforcement learning have emerged as promising alternatives. Liao et al. [40] formulated fence layout as a subset selection problem, while subsequent work [27] proposed a dynamic crowd inflow control system using LSTM networks to capture temporal features and Proximal Policy Optimization (PPO) [41] for real-time entrance flow regulation. Differential Evolution (DE) [42] has also been applied to optimize guardrail layouts and flow guidance in real-world scenarios such as Chengdu East Railway Station, using average travel time and crowd pressure as multi-objective criteria [43].

### B. Macroscopic Crowd Modeling

Macroscopic crowd models treat pedestrian flows as continuous media, offering computational efficiency compared to microscopic approaches. Hughes [44] pioneered the continuum theory for pedestrian flow, introducing the classical model where pedestrians move in the direction of steepest descent of a potential function representing remaining travel time to exits.

Extensions of the Hughes framework have incorporated various control and guidance mechanisms. Anisotropic extensions using metric tensors have been developed to represent channel geometries and directional preferences [45], [46]. These models allow for continuous geometric guidance without explicit prohibition, enabling pedestrians to deviate from preferred directions when necessary.

Constrained Hamilton-Jacobi-Bellman (HJB) formulations have been employed to model strict directional constraints such as one-way corridors. The Bellman optimality principle provides a natural framework for discrete control decisions, unifying anisotropic geometric guidance with directional constraints within a single computational framework.

Despite these advances, existing approaches often treat control parameters as fixed or manually tuned. The bi-level optimization of discrete channel configurations combined with continuous guidance intensities remains underexplored, particularly for large-scale urban pedestrian zones with multiple stages and route choices.

## III. Macroscopic Crowd Dynamics Model

### A. Hughes Model Foundation

We consider a bounded two-dimensional domain $\Omega \subset \mathbb{R}^2$ representing the pedestrian walkway. Let $\rho(x,t)$ denote the crowd density at position $x$ and time $t$, $\phi(x,t)$ the minimal remaining time to reach an exit, and $\mathbf{v}(x,t)$ the mean velocity field. The continuity equation governing mass conservation reads:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \tag{1}$$

In the classical Hughes model [44], pedestrians move in the direction of steepest descent of the potential function:

$$\mathbf{v} = f(\rho)\frac{-\nabla\phi}{|\nabla\phi|}, \tag{2}$$

where $f(\rho)$ is the speed-density relationship. We adopt the Greenshields form:

$$f(\rho) = v_{\max}\left(1 - \frac{\rho}{\rho_{\max}}\right), \tag{3}$$

with $v_{\max}$ the free-flow speed and $\rho_{\max}$ the jam density.

The potential function satisfies the eikonal equation:

$$|\nabla\phi| = \frac{1}{f(\rho)}. \tag{4}$$

While this model captures basic congestion effects, it assumes isotropic movement and cannot represent channel geometries or directional controls. We address these limitations in the following subsections.

### B. Channel Geometric Guidance via Metric Tensor

To represent anisotropic channel geometries such as corridors and barriers, we introduce a spatially varying symmetric positive-definite metric tensor $\mathbf{M}(x)$ that generalizes the classical eikonal equation to anisotropic form [45]:

$$\sqrt{\nabla\phi^\top \mathbf{M}(x)\nabla\phi} = \frac{1}{f(\rho)}. \tag{5}$$

Under this metric, the optimal direction is determined by the tensor-induced steepest descent:

$$\mathbf{v} = f(\rho)\frac{-\mathbf{M}(x)\nabla\phi}{\sqrt{\nabla\phi^\top \mathbf{M}(x)\nabla\phi}}. \tag{6}$$

For a channel region $\Omega_c$ with unit tangent $\tau(x)$ and unit normal $n(x)$, we construct the metric tensor as:

$$\mathbf{M}(x) = \alpha(x)\tau\tau^\top + \beta(x)nn^\top, \quad x \in \Omega_c, \tag{7}$$

where $\alpha(x)$ and $\beta(x)$ control the traversal weights in tangential and normal directions, respectively. Setting $\alpha(x) \gg \beta(x)$ encourages movement along the channel axis while suppressing lateral crossing, achieving continuous geometric guidance without explicit prohibition. Outside channels, $\mathbf{M}(x) = \mathbf{I}$ recovers the isotropic case.

*Remark*: The metric tensor approach represents directional preferences rather than hard constraints. As long as costs remain finite, the model allows individuals to deviate from preferred directions when facing congestion or detour requirements. This motivates the introduction of constrained HJB framework for strict directional controls.

### C. Channel Control via Constrained HJB

Many crowd management measures directly restrict available movement directions rather than merely adjusting traversal costs. Examples include one-way corridors and temporary closures. Such rules are naturally modeled via constrained optimal control.

Let $U(x)$ denote the set of admissible unit directions at position $x$. The velocity satisfies:

$$\mathbf{v} = f(\rho)\mathbf{u}, \quad \mathbf{u}(x) \in U(x), \quad |\mathbf{u}| = 1. \tag{8}$$

By the Bellman optimality principle, the potential function satisfies the constrained HJB equation:

$$\max_{\mathbf{u}\in U(x)} \{-f(\rho)\mathbf{u}\cdot\nabla\phi\} = 1. \tag{9}$$

When $U(x)$ is the full unit circle, (9) reduces to the classical Hughes model. For restricted control sets, various channel rules can be expressed naturally. For a strict one-way channel with allowed direction $\tau(x)$:

$$U(x) = \{\tau(x)\}, \quad x \in \Omega_c, \tag{10}$$

and the HJB equation becomes:

$$-f(\rho)\tau(x) \cdot \nabla\phi = 1, \tag{11}$$

encoding one-way movement as a signed directional constraint.

### D. Unified Discrete Bellman Formulation

To unify anisotropic geometry with discrete constraints, we employ a discrete Bellman formulation. For grid point $x$ and admissible direction $\mathbf{u} \in U(x)$, define the step cost:

$$\text{Cost}(x, \mathbf{u}) = \frac{\Delta x}{f(\rho(x))} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}(x)\mathbf{u}}}, \tag{12}$$

The potential satisfies:

$$\phi(x) = \min_{\mathbf{u} \in U(x)} \left[\phi(x + \Delta x\, \mathbf{u}) + \text{Cost}(x, \mathbf{u})\right]. \tag{13}$$

This formulation couples metric tensor $\mathbf{M}(x)$ for geometric guidance with control set $U(x)$ for directional constraints within a unified Bellman framework. The optimal direction is:

$$\mathbf{u}^*(x) = \arg\min_{\mathbf{u} \in U(x)} \left[\phi(x + \Delta x\, \mathbf{u}) + \text{Cost}(x, \mathbf{u})\right], \tag{14}$$

with velocity $\mathbf{v}(x) = f(\rho)\mathbf{u}^*(x)$.

### E. Multi-Stage Multi-Route Model

Real-world scenarios often involve complex itineraries with multiple stages and route choices. We extend the density field to include stage and route indices, with fixed-probability splitting at decision boundaries.

*1) Stage-Route Decomposition:* Let the complete behavior chain consist of $S$ stages, where stage $s$ has $R_s$ alternative routes. The total density decomposes as:

$$\rho(x, t) = \sum_{s=1}^{S} \sum_{r=1}^{R_s} \rho_{s,r}(x, t), \tag{15}$$

where $\rho_{s,r}(x, t)$ is the density of sub-population in stage $s$ following route $r$.

All sub-populations share the same speed function $f(\rho)$ determined by total density, ensuring collective congestion effects. Sub-populations differ in their potential functions, optimal directions, and exit targets.

*2) Sub-Population Potentials and Velocities:* For each sub-population $(s, r)$, define admissible direction set $U_{s,r}(x)$, metric tensor $\mathbf{M}_{s,r}(x)$, and target regions. The potential $\phi_{s,r}(x)$ satisfies:

$$\begin{aligned}\phi_{s,r}(x) = \min_{\mathbf{u} \in U_{s,r}(x)} \Big[ & \phi_{s,r}(x + \Delta x\, \mathbf{u}) \\ & + \frac{\Delta x}{f(\rho)} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}_{s,r}(x)\mathbf{u}}} \Big].\end{aligned} \tag{16}$$

The optimal direction is:

$$\begin{aligned}\mathbf{u}^*_{s,r}(x) = \arg\min_{\mathbf{u} \in U_{s,r}(x)} \Big[ & \phi_{s,r}(x + \Delta x\, \mathbf{u}) \\ & + \frac{\Delta x}{f(\rho)} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}_{s,r}(x)\mathbf{u}}} \Big],\end{aligned} \tag{17}$$

with velocity $\mathbf{v}_{s,r}(x) = f(\rho)\, \mathbf{u}^*_{s,r}(x)$.

*3) Stage Transitions and Fixed-Probability Splitting:* Sub-populations couple through stage transitions. Let $G_{s,r} \subset \Omega$ be the completion goal region for stage $(s, r)$ with indicator $\chi_{G_{s,r}}(x)$. Upon reaching this region, pedestrians switch to next-stage routes with fixed probabilities.

Let $p_{(s,r)\to q}$ be the fixed probability of switching from $(s, r)$ to $(s + 1, q)$, satisfying:

$$p_{(s,r)\to q} \geq 0, \quad \sum_{q=1}^{R_{s+1}} p_{(s,r)\to q} = 1. \tag{18}$$

The transition term is:

$$Q_{(s,r)\to(s+1,q)}(x, t) = p_{(s,r)\to q}\, \kappa_{s,r}\, \chi_{G_{s,r}}(x)\, \rho_{s,r}(x, t), \tag{19}$$

where $\kappa_{s,r} > 0$ is the stage switching rate.

The density evolution for each sub-population includes advection and transition source terms:

$$\frac{\partial \rho_{s,r}}{\partial t} + \nabla \cdot (\rho_{s,r}\mathbf{v}_{s,r}) = Q^{\text{in}}_{s,r} - Q^{\text{out}}_{s,r}, \tag{20}$$

where $Q^{\text{in}}_{s,r}$ is inflow from the previous stage and $Q^{\text{out}}_{s,r}$ is outflow to the next stage.

## IV. Numerical Solution

We discretize the domain using a uniform two-dimensional grid, with the control direction set represented as an 8-neighborhood basis with masking for regional restrictions.

**Unified Bellman Solver.** Rather than separate eikonal and HJB solvers, we use a unified discrete Bellman shortest-path update. For candidate direction $\mathbf{u}$, precompute geometric step factors from the local tensor, then combine with the speed function to form the complete step cost. A priority queue propagates potential values outward from exits until convergence. When the control set is unrestricted, this procedure is equivalent to monotone solution of the anisotropic eikonal equation; when controls are restricted, it naturally reduces to constrained HJB solution, covering all required scenarios within a unified framework.

**Density Update.** After obtaining converged potentials, recover velocity fields using (17), substitute into the continuity equation, and advance density explicitly using an upwind flux scheme:

$$\rho^{n+1} = \rho^n - \Delta t\, \nabla \cdot (\rho^n \mathbf{v}^n). \tag{21}$$

The time step satisfies the CFL stability condition.

**Evaluation Oracle.** For any given control configuration $z = (s, \eta)$, the simulation workflow constitutes a lower-level evaluation operator:

$$(\rho, \phi, \mathbf{v}) = \mathcal{S}(z; \hat{p}), \tag{22}$$

where $\hat{p}$ represents fixed route preference parameters identified from historical data. This operator provides the foundation for subsequent control optimization.

## V. Control Evaluation Metrics

We distinguish between behavior validation metrics (for verifying model mechanisms) and control evaluation metrics (for assessing policy impacts). This section focuses on the latter.

### A. Efficiency Metric: Total Travel Time

For simulation horizon $[0, T]$, the total travel time metric is:

$$J_1(z \mid \hat{p}) = \int_0^T \int_\Omega \rho(x, t)\, dx\, dt. \tag{23}$$

This measures aggregate dwell time of the crowd, providing a stable basis for comparing different control strategies.

### B. Safety Metric: High-Density Exposure Time

Let $\rho_{\text{safe}}$ be the safe density threshold. The exposure time metric is:

$$J_2(z \mid \hat{p}) = \int_0^T \int_\Omega \mathbf{1}_{\rho(x,t) > \rho_{\text{safe}}}\, dx\, dt. \tag{24}$$

This captures cumulative space-time exposure to dangerous crowding conditions, reflecting management concerns about sustained congestion risks.

### C. Balance Metric: Channel Flow Variance

Let $q_c(t)$ be the flux through channel $c$ at time $t$, with cumulative flow $F_c = \int_0^T q_c(t)\, dt$. The load balancing metric is:

$$J_5(z \mid \hat{p}) = \text{Var}(F_1, \ldots, F_m). \tag{25}$$

Lower variance indicates more even distribution of traffic across channels, reducing risk concentration and improving facility utilization.

### D. Dimensionless Normalization

Because $J_1$, $J_2$, and $J_5$ have different physical units and numerical scales, we further define normalized indicators for policy comparison and optimization. Let $M_{\text{tot}} = M_{\text{init}} + M_{\text{in}}$ denote the total evaluated crowd mass, let $|\Omega_w|$ be the walkable area, and let $F = \sum_{c=1}^m F_c$ be the total cumulative channel flow. We define

$$\tilde{J}_1(z \mid \hat{p}) = \frac{J_1(z \mid \hat{p})}{M_{\text{tot}} T}, \tag{26}$$

$$\tilde{J}_2(z \mid \hat{p}) = \frac{J_2(z \mid \hat{p})}{|\Omega_w| T}, \tag{27}$$

and

$$\tilde{J}_5(z \mid \hat{p}) = \frac{\text{Var}(F_1, \ldots, F_m)}{F^2(m-1)/m^2} = \frac{m^2}{m-1} \text{Var}(p_1, \ldots, p_m), \tag{28}$$

where $p_c = F_c / F$. The normalized metrics are dimensionless, with $\tilde{J}_2 \in [0, 1]$ and $\tilde{J}_5 \in [0, 1]$.

### E. Scalarized Optimization Objective

Given weights $(\lambda_1, \lambda_2, \lambda_5)$, the scalarized objective is:

$$J(z \mid \hat{p}) = \lambda_1 \tilde{J}_1(z \mid \hat{p}) + \lambda_2 \tilde{J}_2(z \mid \hat{p}) + \lambda_5 \tilde{J}_5(z \mid \hat{p}), \tag{29}$$

with the optimization problem:

$$\min_{z \in \mathcal{Z}} J(z \mid \hat{p}). \tag{30}$$

Our algorithm solves this scalarized problem for fixed weights, rather than directly generating the full Pareto frontier. Approximate Pareto sets can be obtained by running with different weight combinations.

## VI. Optimization Method

### A. Control Variable Definition

We distinguish controllable variables from exogenous behavioral parameters:

**Controllable variables**:

1) *Channel direction configuration* $s = (s_1, \ldots, s_C)$, where $s_c \in \{-1, 0, +1\}$ indicates reverse one-way, bidirectional, or forward one-way for channel $c$.
2) *Geometric guidance intensity* $\eta = (\eta_1, \ldots, \eta_C)$, where $\eta_c = \alpha_c / \beta_c \geq 1$ is the anisotropy ratio.

Thus the optimization variable is $z = (s, \eta)$.

**Exogenous parameters**: Route preference parameters $\hat{p}$ are identified from historical data and fixed during optimization. Decision region locations are reserved for future extension.

### B. Structure-Aware Hybrid Block Optimization

Treating this as a black-box optimization ignores three key structures: mixed discrete-continuous variables, local Bellman response characteristics, and the expensive multi-group simulation structure.

We propose Structure-Aware Hybrid Block Optimization (SA-HBO), with four core components:

1) Discrete block neighborhood search for channel directions;
2) Surrogate pre-screening to reduce full simulations;
3) Bellman warm-start with local re-solving for efficient evaluation;
4) Continuous block projected stochastic approximation for guidance intensities.

### C. Discrete Block Update

Fixing continuous variables $\eta^k$, the discrete subproblem at iteration $k$ is:

$$s^{k+1/2} \approx \arg\min_{s\in\mathcal{S}} J(s, \eta^k \mid \hat{p}). \tag{31}$$

Define the neighborhood of current configuration $s^k$ as:

$$\mathcal{N}(s^k) = \left\{ s : \|s - s^k\|_0 \le \nu \right\}, \tag{32}$$

where $\nu$ is the maximum number of simultaneously flipped channels (typically 1 or 2).

*Surrogate pre-screening*: To reduce full simulations, construct a local surrogate for each candidate $s' \in \mathcal{N}(s^k)$:

$$\widehat{J}(s', \eta^k) = \mu_1 \widehat{J}_1^{\text{local}} + \mu_2 \widehat{J}_2^{\text{local}} + \mu_3 \widehat{J}_3^{\text{flow}}, \tag{33}$$

where the local metrics approximate density changes, exposure changes, and flow redistribution. Select top $K$ candidates by surrogate value for full PDE simulation.

### D. Continuous Block Update

Fixing discrete variables $s^{k+1/2}$, solve for $\eta$:

$$\eta^{k+1} \approx \arg\min_{\eta\in\mathcal{H}} J(s^{k+1/2}, \eta \mid \hat{p}), \tag{34}$$

with feasible set $\mathcal{H} = \{\eta : \eta_c \ge 1\}$.

Since gradients are unavailable analytically, we use projected stochastic approximation. Generate Rademacher perturbation $\Delta_\eta^k \in \{-1, +1\}^C$, construct two-sided perturbations $\eta_\pm^k = \eta^k \pm c_k \Delta_\eta^k$, and project to feasible set. The stochastic gradient estimate is:

$$\widehat{g}_\eta^k = \frac{J(s^{k+1/2}, \tilde{\eta}_+^k \mid \hat{p}) - J(s^{k+1/2}, \tilde{\eta}_-^k \mid \hat{p})}{2c_k} \odot (\Delta_\eta^k)^{-1}. \tag{35}$$

The candidate update is:

$$\eta_{\text{cand}}^{k+1} = \Pi_{\mathcal{H}} \left( \eta^k - \alpha_k D_\eta^{-1} \widehat{g}_\eta^k \right), \tag{36}$$

with diagonal preconditioner $D_\eta$.

*Acceptance/Rejection*: Compare candidate objective $J_{\text{cand}}$ with current $J_{\text{cur}}$. Accept if $J_{\text{cand}} \le J_{\text{cur}} - \varepsilon_{\text{acc}}$; otherwise reject and optionally backtrack. Maintain best solution $z_{\text{best}}$ throughout.

### E. Bellman Warm-Start and Local Re-solving

Control changes typically affect only local regions. We identify the affected region $\Omega_{\text{chg}}^k$ where $U_{s,r}(x; \tilde{z}) \ne U_{s,r}(x; z^k)$ or $\mathbf{M}_{s,r}(x; \tilde{z}) \ne \mathbf{M}_{s,r}(x; z^k)$. Using the old potential as initialization, iterate only within an expanded neighborhood $\Omega_{\text{chg}}^{k,\delta}$ until convergence.

This accelerates candidate evaluation. Periodic global corrections prevent error accumulation.

### F. Algorithm Summary

1) Initialize potentials and density, compute $J(z^0 \mid \hat{p})$, set $z_{\text{best}} = z^0$.
2) For each iteration:
   - *Discrete update*: Generate neighborhood, screen with surrogate, evaluate top candidates with Bellman warm-start, select $s^{k+1/2}$.
   - *Continuous update*: Generate perturbations, compute stochastic gradient, generate candidate, accept/reject to obtain $\eta^{k+1}$.
   - *State update*: Set $z^{k+1} = (s^{k+1/2}, \eta^{k+1})$, compute objective, update $z_{\text{best}}$ if improved.
3) Return $z^\star = z_{\text{best}}$.

Stopping criteria include: small objective change, no discrete variable change for consecutive iterations, or maximum iteration/simulation/time limits.

## VII. Numerical Experiments

This section presents numerical experiments to validate the model mechanisms, evaluate control strategies, and demonstrate the optimization framework. The experimental design follows a two-layer validation logic: first verifying whether the model mechanisms authentically alter local crowd behaviors, then comparing different channel control strategies in terms of efficiency, safety, and load balancing.

### A. Experimental Setup

*1) Three-Channel Scenario:* The experiments employ a three-channel scenario as the primary testbed for mechanism validation and strategy comparison. The geometry consists of an initial crowd region on the left, an exit region on the right, and three passages (upper, middle, lower) through a central wall separating the two regions.

Numerical parameters are set as follows: grid resolution $96 \times 72$, spatial step $\Delta x = 0.5$, free-flow speed $v_{\max} = 1.5$, maximum density $\rho_{\max} = 5.0$, initial density $\rho_{\text{init}} = 2.2$, CFL coefficient $0.9$, maximum simulation steps $600$, and time horizon $T = 40$.

*2) Evaluation Metrics:* The experiments employ three primary raw metrics and their normalized counterparts:

- *Efficiency metric* $J_1$ and $\tilde{J}_1$: total travel time and its mass-time normalized form;
- *Safety metric* $J_2$ and $\tilde{J}_2$: high-density exposure time and its space-time normalized form;
- *Balance metric* $J_5$ and $\tilde{J}_5$: channel flow variance and its normalized imbalance index.

Behavioral indicators include channel capture domain ratios, flow distribution ratios, direction consistency indices, and approach angle distributions.

### B. Mechanism Validation

The mechanism validation experiments address two fundamental questions: (1) Do geometric guidance $M(x)$ and directional constraints $U(x)$ authentically alter local decision-making and flow patterns? (2) Does the model produce sufficiently sensitive and interpretable responses to configuration changes?

TABLE I
BIDIRECTIONAL $U(x)$ VALIDATION RESULTS

| Configuration | Westbound Share | Counter-flow Mass | Head-on Exposure |
|---|---|---|---|
| Bidirectional baseline | 73.19% | 152.15 | 1055.74 |
| Middle channel one-way | 0 | 0 | 0 |

TABLE II
CONFIGURATION SENSITIVITY RESULTS

| Config | Dominant Capture | Dominant Flow | Upper | Middle | Lower |
|---|---|---|---|---|---|
| middle | middle | middle | ∼1/3 | dominant | ∼1/3 |
| top | top | top | dominant | ∼1/3 | ∼1/3 |
| bottom | bottom | bottom | ∼1/3 | ∼1/3 | dominant |

*1) Effect of Geometric Guidance $M(x)$:* To verify whether geometric guidance alters approach patterns and attraction domains, we compare the baseline configuration (neither $M$ nor $U$) against the $M$-only configuration (anisotropic guidance applied to the middle channel).

**Results:** Under the baseline configuration, the three channels exhibit approximately balanced capture domains: upper $33.73\%$, middle $30.00\%$, lower $34.98\%$. Flow distributions are similarly balanced: upper $33.82\%$, middle $32.35\%$, lower $33.83\%$.

Under the $M$-only configuration, the middle channel capture domain expands significantly to $63.41\%$, and its flow share jumps to $99.32\%$. This demonstrates that geometric guidance $M(x)$ reshapes local attraction domains and fundamentally alters approach patterns.

*2) Effect of Directional Constraints $U(x)$:* To verify whether directional constraints alter local optimal directions and channel selection, we compare the baseline against the $U$-only configuration (middle channel restricted to eastward-only flow).

**Results:** Under the $U$-only configuration, the middle channel capture domain drops to $0$, and its flow share drops to $0$, with upper and lower channels splitting traffic approximately $50/50$. This confirms that $U(x)$ significantly rewrites feasible path choices.

To further verify the suppression of counter-flow and head-on conflicts, we conduct a bidirectional flow sub-experiment. Two configurations are tested: bidirectional baseline (westbound and eastbound crowds simultaneously) and bidirectional with middle channel one-way rule.

The results in Table I show that when the middle channel enforces a one-way rule, westbound crowds are completely excluded, and counter-flow mass time and head-on cell exposure are both driven to zero. This confirms that $U(x)$ not only changes path selection but effectively suppresses counter-flow and head-on conflicts under bidirectional demand.

*3) Combined $U+M$ Effects and Configuration Sensitivity:* To verify whether the combined $U+M$ model produces distinguishable responses to different channel configurations, we test three setups: middle-channel-controlled ($U+M$-middle), upper-channel-controlled ($U+M$-top), and lower-channel-controlled ($U+M$-bottom).

Table II demonstrates that the $U+M$ model exhibits stable and interpretable responses to configuration changes: when the middle channel is controlled, it becomes the dominant capture and flow channel; when the upper or lower channel is controlled, the corresponding channel becomes dominant. This sensitivity validates the model as a foundation for subsequent strategy comparison and optimization.

*4) Mechanism Validation Conclusions:* The experimental results support four key conclusions:

1) Geometric guidance $M(x)$ significantly alters local approach patterns and channel attraction domains, enabling earlier alignment with channel axes and smoother convergence.
2) Directional constraints $U(x)$ reshape feasible paths and directional organization, effectively suppressing counter-flow and head-on conflicts under bidirectional demand.
3) The combined $U+M$ model produces configuration-sensitive responses that are both rule-compliant and behaviorally realistic.
4) Either $U$ or $M$ alone yields incomplete or behaviorally unrealistic responses; the full model is necessary for reliable optimization.

### C. Control Strategy Comparison

This subsection compares different channel control strategies in terms of normalized efficiency ($\tilde{J}_1$), normalized safety ($\tilde{J}_2$), and normalized load balancing ($\tilde{J}_5$). The strategies under comparison include:

- Baseline (no control);
- Upper-channel-guided;
- Middle-channel-guided;
- Lower-channel-guided.

Detailed results and analysis will be presented in the full version of this paper.

### D. Multi-Stage Flow Validation Experiments

This subsection validates the multi-stage behavior chain and probabilistic splitting mechanism. The experiments compare single-stage approximations against full multi-stage models to demonstrate how stage transitions and fixed-probability splitting alter density evolution and exit load distribution.

Detailed results and analysis will be presented in the full version of this paper.

### E. Optimal Control Parameter Search

This subsection presents the first-version automated parameter search results. The search enumerates discrete strategy variables, performs coarse grid search on guidance intensity $\eta$, and searches over fixed splitting probability candidates to identify optimal control configurations under different objective weight combinations $(\lambda_1, \lambda_2, \lambda_5)$.

Detailed results and analysis will be presented in the full version of this paper.

## VIII. CONCLUSION

This paper presents a bi-level optimization framework for crowd channel management that combines anisotropic geometric guidance with discrete directional constraints. The proposed SA-HBO algorithm exploits problem structure through

surrogate pre-screening, Bellman warm-start, and block coordinate updates to handle the expensive simulation-based optimization efficiently.

The numerical experiments validate that: (1) geometric guidance $M(x)$ and directional constraints $U(x)$ authentically alter local crowd behaviors; (2) the combined model exhibits stable sensitivity to configuration changes; (3) directional constraints effectively suppress counter-flow and head-on conflicts under bidirectional demand. These findings establish the foundation for subsequent strategy optimization studies.

Future work will extend to full multi-objective optimization, sensitivity analysis under demand perturbations, and calibration against real-world observation data.

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